

Owls

PHYLUM	Chordata
CLASS	Aves
ORDER	Strigiformes
FAMILIES	2
SPECIES	242

Often heard yet seldom seen, owls are hunters that operate mainly at night. They resemble the day-flying predators, hawks and falcons, in having sharp talons and hooked bills for catching and subduing prey, but in addition owls have several adaptations that help them hunt in the dark. Their eyes are very large, to gather all available light, and are forward-facing, to help them judge distance. They have exceptionally acute hearing, and their soft plumage enables them to fly silently. The 2 families of owls—typical owls and barn owls—are found worldwide in most habitats, from dense forest to tundra.

Reproduction

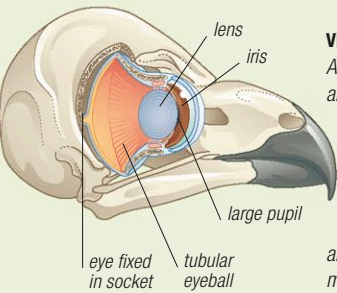
Owls do not build their own nest. Instead, they rely on the previous years' efforts of other birds or simply select a suitable cavity in the ground, a tree, a rocky crevice, or a building. The almost spherical eggs are laid in the nest or straight onto the surface. Most owls lay 2–7 eggs, which usually hatch at 2-day intervals, leading to large discrepancies in the age of the chicks in a brood. If food is scarce, the older chicks take the larger part of the food offered, while the younger chicks may starve.



YOUNG
The youngest owl in a brood (here, snowy owls) can be 2 weeks younger than the oldest.

Anatomy

Owls are highly distinctive, with an upright posture, a large, rounded head, and a short tail. The outer toe is reversible, allowing it to point forward or backward, improving the ability to perch or grab prey. Owls have excellent eyesight, which works as well in daylight as it does at night. They also have exceptional hearing, easily picking up the faint rustle made by a small mammal, even under snow. Some species can hunt in total darkness, their asymmetrical ear openings giving them a three-dimensional perception of sound. All owls have soft, dense plumage, with soft fringes on their flight feathers that muffle the sound of air turbulence.



VISION
An owl's eyeballs are tubular and cannot be swiveled. As a result, if the bird needs to look to the side, it must move its whole head. To compensate, an owl can rotate its head and neck about an arc of more than 270 degrees.

Feeding

Owls take a wide variety of living prey, the size of which depends on the owl species. Most feed on insects, birds, or small mammals, and large owls quite commonly catch and eat smaller owls. Owls that live in woods and forests tend to drop from a stationary position onto their prey, but those that occur in open country must hunt on the wing, in slow, low-level, quartering flight.

NOCTURNAL HUNTER

Under cover of darkness, a little owl closes silently upon its prey. Equipped with sharp talons, acute eyesight, and exceptional hearing, an owl is a formidable nighttime predator.

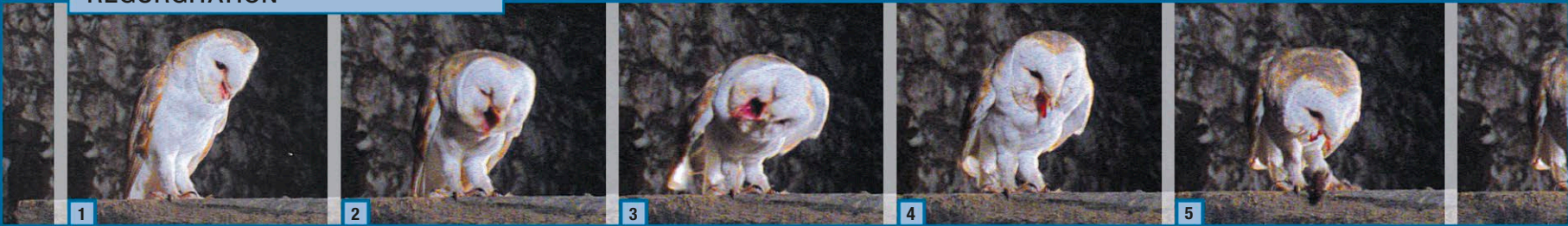
FISHING

Several species of owls, including Pel's fishing owl (right), feed on fish. They swoop down to pluck fish from the water surface, striking them with their talons and then taking hold with their bristly foot pads.



PELLETS
Owls generally swallow their food whole, taking in fur, feathers, bones, and insect chitin. Later, they regurgitate the indigestible parts (see below) in the form of compact pellets, which collect beneath their nest or roosting site.

REGURGITATION



SWALLOWING PREY
A barn owl swallows its prey whole, including much indigestible matter, such as fur and bones.

PREPARING TO CAST
About 6 or 7 hours after eating, the owl is about to regurgitate (or cast) the indigestible parts of its meal.

ROUNDED PELLETS
To ease its passage, the undigested food is rolled into a pellet, with hard parts wrapped in fur and feathers.

CAUGHT IN THE THROAT
Despite being rounded, the pellet causes the owl to strain as it moves up the esophagus.

RELEASING THE PELLET
Finally, a shiny black pellet is cast from the mouth. The owl may cast a second one within 24 hours.